1.Build Array From Permutation:

```
class Solution {
    public int[] buildArray(int[] nums) {
        int n = nums.length;
        for (int i = 0; i < n; i++) {
            int newVal = nums[nums[i]] % n;
            nums[i] += newVal * n;
        }
        for (int i = 0; i < n; i++) {
            nums[i] /= n;
        }
        return nums;
    }
}
```

2.Shuffle the Array

```
class Solution {
    public int[] shuffle(int[] nums, int n) {
        int len = nums.length;
        int[] ans = new int[len];
        for (int i = 0; i < len; i++) {
            if (i % 2 == 0) {
                // Even positions get elements from the first half
                ans[i] = nums[i / 2];
            } else {
                // Odd positions get elements from the second half (starting at index n)
                ans[i] = nums[n + (i / 2)];
            }
        }
        return ans;
    }
}
```

3.Find the highest Altitude

```
class Solution {
    public int largestAltitude(int[] gain) {
        int maxAltitude = 0;
        int currentAltitude = 0;

        for (int i = 0; i < gain.length; i++) {
            currentAltitude += gain[i];
```

```
            maxAltitude = Math.max(maxAltitude, currentAltitude);
        }

        return maxAltitude;
    }
}
```